

SOE WAI YAN

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EDUCATION

Massachusetts Institute of Technology (MIT)

May 2028

B.S. in Electrical Engineering and Computer Science GPA: 4.5/5.0

Cambridge, MA

- **Relevant Coursework:** Introduction to Algorithms, Signal Processing, Electric Circuits, Introduction to C in Assembly, Robotics Science and Systems, Introduction to Autonomous Machines I & II

SKILLS

Programming Languages: Python, C++, C, Assembly

Robotics & ML: ROS 2, PyTorch, OpenCV, CUDA, Computer Vision, Imitation Learning, Motion Planning (A*/RRT), Visual Servoing, Isaac Sim

Hardware: Circuit Design, Electronic Prototyping, Arduino, Jetson Nano, ESP32, LiDAR, CAN bus

Tools: Linux, MATLAB, AutoCAD, Fusion 360

EXPERIENCE

MIT SPARK Lab (Robotics Lab)

June 2026 – Present

Undergraduate Researcher ROS 2, Python, PyTorch

Cambridge, MA

- Trained vision-based manipulation policies (ACT, $\pi 0.5$ vision-language-action model) to autonomously grasp objects with a real 7-DoF bimanual robot arm (AgileX Nero ALOHA) from 3-camera RGB input
- Collected 300+ teleoperated demonstrations and built the full data pipeline — raw HDF5 capture → normalization → LeRobot format → LoRA fine-tuning of $\pi 0.5$ on dual RTX 4090 GPUs
- Deployed policies for real-time inference on an edge GPU (RTX 4060) and iteratively improved grasp performance through on-hardware debugging and policy tuning

MIT Computer Science and Artificial Intelligence Laboratory (CSAIL)

Sept 2025 – Dec 2025

Undergraduate Researcher

Cambridge, MA

- Designed and prototyped modular robot hardware, integrating sensors, actuators, and microcontrollers via 3D printing, electronics simulation, and embedded control
- Fabricated and tested reconfigurable hardware prototypes for robot morphology optimization
- Developed a lightweight UI to visualize design configurations and control mappings

Core Research Program, Georgia State University

June 2024 – May 2025

Undergraduate Researcher

Atlanta, GA

- Programmed the Arduino IDE to calibrate an Adafruit AS7341 multispectral sensor for soil nutrient detection
- Designed and assembled the sensor hardware and communication interface for an autonomous soil-analysis system
- Co-authored a research paper presented at the NCUR 2025 conference (Pittsburgh, PA)

PROJECTS

Autonomous Elevator-Button-Press System

June 2026 – Present

ROS 2, Computer Vision, Perception, Manipulation

- Built an end-to-end vision-guided pipeline enabling a 7-DoF bimanual robot arm (AgileX Nero ALOHA) to autonomously locate and press elevator call buttons, from RGB-D detection to physical actuation
- Developed a real-time 2D button detector with temporal smoothing, then backprojected pixels to 3D using depth and a calibrated camera-to-arm extrinsic transform
- Implemented Cartesian motion execution over ROS 2 with TCP-offset calibration and pose-feedback verification; diagnosed and resolved low-level hardware faults (CAN-bus tuning, motor brake-release timing, fault recovery)

Autonomous Racecar Perception & Control (Team of 5)

Feb 2026 – March 2026

Python, C++, ROS 2, OpenCV

- Built a camera-based visual servoing pipeline (HSV segmentation, SIFT, YOLO, homography ground-plane projection) for real-time detection and localization, reaching 0.79 median IoU and 1.5 cm mean positioning error
- Designed a LiDAR wall-following controller (PD control + RANSAC line fitting) and an independent $>180^\circ$ -FOV safety controller, achieving stable 0.55–0.71 m wall tracking at 1–2 m/s with sub-0.25 m emergency braking
- Implemented and benchmarked A* and RRT planners over an occupancy grid and a pure-pursuit controller with off-path recovery, completing the 400 m race lane at a 95% success rate on a physical autonomous racecar

School Bus Tracking App (BUSMATE) (Team of 4)

Aug 2024 – May 2025

IoT, GPS, Figma

- Developed an affordable IoT-based school bus tracking prototype using a low-cost GPS sensor; led user research and prototype design in Figma
- Won 1st Place in the 2024 Perimeter College Innovative Challenge Competition